

Alaric D'Souza

Infectious Diseases Fellow

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About me

Alaric is a post-doctoral fellow at the Harvard School of Public Health and an Infectious Diseases Fellow at Boston Children's Hospital. He cares for pediatric patients at Boston Children's Hospital, Brigham and Women's Hospital, and Beth Israel Deaconess Medical Center. He did his Pediatrics Residency at Boston Children's Hospital. He also completed his MD/PhD at Washington University in St. Louis, and his Computational and Systems Biology dissertation research on Microbial Ecology and antibiotic resistance genes. Alaric hopes to lead his own research group studying microbial ecology, infectious disease epidemiology, and antibiotic resistance.

Skills

Software Packages
Adobe Illustrator, Adobe Photoshop, EndNote, LaTeX, Microsoft Office

Programming Languages
Python, R, Bash, Wolfram Mathematica

Languages
English (native)
Hindi (beginner)

Misc.
Computer Hardware, Machining and Fabrication, Welding

research interests

Microbial ecology, bacterial genomics, and antibiotic resistance.

education

2014-22 MD — Washington University in St. Louis

2014-22 PhD (defense date: 3/30/2020) — Dantas Lab, Washington University in St. Louis
Computational and Systems Biology

2010-14 BS — Yale University
Ecology and Evolutionary Biology

research and work experience

2025-pres. Grad Lab — Harvard School of Public Health
Post-doctoral Fellow | Antibiotic use trends and target trial emulation with large scale electronic health records data

2024-pres. Infectious Diseases Fellowship — Boston Children's Hospital
Clinical Fellow | Infectious Diseases care for pediatric patients at Boston Children's Hospital, Brigham and Women's Hospital, and Beth Israel Deaconess Medical Center.

2022-24 Pediatrics Residency — Boston Combined Residency Program
Resident Physician *(Accelerated Research Pathway)* | Medical care for pediatric patients at Boston Children's Hospital, Brigham and Women's Hospital, and Boston Medical Center.

2016-20 Dantas Lab — Washington University in St. Louis
Doctoral Candidate | Ecology of hospital built environment bacteria, international travelers, and infant gut microbiota.

2015 Dantas Lab — Washington University in St. Louis
Rotation Student | Autologous Fecal Microbial Transplant Analysis

2012-14 Powell Lab — Yale University
Undergraduate Researcher | Foraging behavior of *Aedes aegypti* mosquitoes.

2013 Gagneux Lab — Swiss Tropical and Public Health Institute
Undergraduate Researcher | Antibiotic resistance and mutation rates in *Mycobacterium tuberculosis*.

2012 Schlesinger Lab — The Ohio State University
Undergraduate Researcher | Macrophage receptor signaling for *Mycobacterium tuberculosis* phagosome lysosome fusion escape.

2011 Seva Dhan — Mumbai, India
Summer Intern | Treatment of alcoholics and drug addicts.

2011 Course Consulting — Canyon High School
Consultant | Course design for AP Psychology and IB Theory of Knowledge.

2009 Mentee of Dr. P.S. Wijewarnasuriya — Army Research Labs
Summer Intern | Semiconductor photoresist optimization for infrared detectors.

publications

* authors contributed equally | ** corresponding author

2026 Microbiome-antibiotic-immune interactions influence mortality in preterm infant gut microbiota-humanized gnotobiotic mice
Schwartz DJ**, Wong GG, Wardenburg KE, Erickson IR, Tung J, Ning J, Zhao L, Shalon N, D'Souza AW, Ferreiro A, Silverstein RB, Crofts TS, Hall-Moore C, Wallace MA, Robinson JI, Henderson JP, Burnham CAD, Warner BB, Tarr PI, Dantas G**
Cell Reports Medicine | 10.1016/j.xcrm.2026.103161

2026 Fatal Neuroinvasive Astrovirus VA1 Encephalitis in a Child With High-Risk B-ALL Receiving Blinatumomab
D'Souza AW, Miozzo P, Wilson-Murphy M, Place AE, Dalai S, Miller S, Levy O, Rakoff-Nahoum S**
Pediatric Blood and Cancer | 10.1002/1545-5017.70630

2026 Case Report: *Pseudomonas* liver abscess in a previously healthy child with homozygous pathogenic S allele variant of the *SERPINA1* gene.
Thomas S**, D'Souza AW, Alhezam M, Platt CD**, Levy O**
Frontiers in Immunology | 10.3389/fimmu.2026.1755963

2025 Linezolid-resistant and vancomycin-resistant *Enterococcus faecium* urinary isolate in a pediatric B-ALL patient
Seevak E, Sharma TS, D'Souza AW**
ASM Case Reports | 10.1128/asmcr.00085-25

2023 Daily occupational exposure in swine farm alters human skin microbiota and antibiotic resistome
Chen DR, Cheng K, Wan L, Li G, Zhao DH, Yu Y, Liao XP, Liu YH, D'Souza AW**, Lian XL**, Sun J**
iMeta | 10.1002/imt2.158

2022 Acute and persistent effects of commonly-used antibiotics on the gut microbiome and resistome in healthy adults
Anthony WE, Wang B, Sukhum KV, D'Souza AW, Hink T, Cass C, Seiler S, Reske KA, Coon C, Dubberke ER, Burnham CA, Dantas G, Kwon JH
Cell Reports | 10.1016/j.celrep.2022.110649

2022 Structural and molecular rationale for the diversification of resistance mediated by the Antibiotic_NAT family
Stogios PJ, Bordeleau E, Xu Z, Skarina T, Evdokimova E, Chou S, Diorio-Toth L, D'Souza AW, Patel S, Dantas G, Wright GD, Savchenko A
Communications Biology | 10.1038/s42003-022-03219-w

2021 Destination shapes antibiotic resistance gene acquisitions, abundance increases, and diversity changes in Dutch travelers.
D'Souza AW*, Boolchandani M*, Patel S, Galazzo G, Hattem J, Arcilla M, Melles D, Jong M, Schultsz C, Dantas G, Penders J
Genome Medicine | 10.1186/s13073-021-00893-z

2021 Manure microbial communities and resistance profiles reconfigure after transition to manure pits and differ from fertilized field soil
Sukhum K, Vargas R, D'Souza AW, Boolchandani M, Patel S, Kesaraju A, Walljasper G, Hedge H, Ye Z, Valenzuela R, Gunderson P, Bendixsen C, Dantas G, Shukla S
mBio | 10.1128/mBio.00798-21

2020 Environmental remodeling of human gut microbiota and antibiotic resistome in livestock farms
Sun J*, Liao XP*, D'Souza AW*, Boolchandani M*, Li SH*, Cheng K, Martinez JL, Li L, Feng YJ, Fang LX, Huang T, Xia J, Yu Y, Zhou YF, Sun YX, Deng XB, Zeng ZL, Jiang HX, Fang BH, Tang YZ, Lian XL, Zhang RM, Fang ZW, Yan QL, Dantas G, Liu YH
Nature Communications | 10.1038/s41467-020-15222-y

2019 Genomic characterization of antibiotic resistant *Escherichia coli* isolated from domestic chickens in Pakistan
Rafique M*, Potter RF*, Ferreiro A, Wallace MA, Rahim A, Malik AA, Siddique N, Abbas MA, D'Souza AW, Burnham CD, Ali N, Dantas G
Frontiers in Microbiology | 10.3389/fmicb.2019.03052

2019 Cotrimoxazole prophylaxis increases resistance gene prevalence and α-diversity but decreases β-diversity in the gut microbiome of HIV-exposed, uninfected infants.
D'Souza AW*, Moodley-Govender E*, Berla B, Kelkar T, Wang B, Sun X, Daniels B, Coutsoudis A, Trehan I, Dantas G
Clinical Infectious Diseases | 10.1093/cid/ciz1186

2019 Breakpoint beware: reliance on historical breakpoints for Enterobacteriaceae leads to discrepancies in interpretation of susceptibility testing for carbapenems and cephalosporins and gaps in detection of carbapenem-resistant organisms.
Yarbrough ML, Wallace MA, Potter RF, D'Souza AW, Dantas G, Burnham CD
European Journal of Clinical Microbiology & Infectious Diseases | 10.1007/s10096-019-03711-y

2019 Spatiotemporal dynamics of multidrug resistant bacteria on intensive care unit surfaces.
D'Souza AW*, Potter RF*, Wallace M, Shupe A, Patel S, Sun X, Gul D, Kwon JH, Andleeb S, Burnham CD, Dantas G
Nature Communications | 10.1038/s41467-019-12563-1

2019 Phenotypic and genotypic characterization of linezolid resistant *Enterococcus faecium* from the USA and Pakistan.
Wardenburg K*, Potter R*, D'Souza AW, Hussain T, Wallace M, Andleeb A, Burnham CA, Dantas G
Journal of Antimicrobial Chemotherapy | 10.1093/jac/dkz367

2019 Sequencing-based methods and resources to study antimicrobial resistance.
Boolchandani M*, D'Souza AW*, Dantas G
Nature Reviews Genetics | 10.1038/s41576-019-0108-4

2018 Infant diet and maternal gestational weight gain predict early metabolic maturation of gut microbiomes.
Baumann-Dudenhoeffer AM, D'Souza AW, Tarr PI, Warner BB, Dantas G
Nature Medicine | 10.1038/s41591-018-0216-2

2018 *Superficieibacter electus* gen. nov., sp. nov., an Extended-Spectrum β-Lactamase Possessing Member of the Enterobacteriaceae Family, Isolated From Intensive Care Unit Surfaces.
Potter RF*, D'Souza AW*, Wallace MA, Shupe A, Patel S, Gul D, Kwon JH, Beatty W, Andleeb S, Burnham CD, Dantas G
Frontiers in Microbiology | 10.3389/fmicb.2018.01629

2017 Draft Genome Sequence of the *bla*OXA-436- and *bla*NDM-1-Harboring *Shewanella putrefaciens* SA70 Isolate.
Potter RF*, D'Souza AW*, Wallace MA, Shupe A, Patel S, Gul D, Kwon JH, Andleeb S, Burnham CD, Dantas G
Genome Announcements | 10.1128/genomeA.00644-17

2016 The rapid spread of carbapenem-resistant Enterobacteriaceae.
Potter RF*, D'Souza AW*, Dantas G
Drug Resistance Updates | 10.1016/j.drup.2016.09.002

2014 Malignant cancer and invasive placentation: A case for positive pleiotropy between endometrial and malignancy phenotypes.
D'Souza AW and Wagner GP
Evolution, Medicine, and Public Health | 10.1093/emph/eou022

published abstracts

2021 Microbiome and immune disruption accompany mouse death in a gnotobiotic mouse model of neonatal sepsis
Schwartz D, Wardenburg K, Shalon N, Ning J, Crofts T, D'Souza A, Robinson J, Henderson J, Warner B, Tarr P, Dantas G
Journal of the Pediatric Infectious Diseases Society | 10.1093/jpids/piab031.010

2020 The Gut Microbiome and Resistome of Healthy Volunteers are Restructured After Short Courses of Antibiotics
Anthony W, Sukhum K, Cass C, Reske K, Seiler S, Hink T, Coon C, D'Souza A, Wang B, Sun S, Dubberke E, Burnham C-A, Dantas G, Kwon JH
Infection Control & Hospital Epidemiology | 10.1017/ice.2020.476

2017 Longitudinal Analysis of ICU Surface Multidrug-resistant Organism Contamination in the US and Pakistan
Kwon JH, D'Souza A, Potter R, Wallace M, Shupe A, Patel S, Gul D, Andleeb S, Burnham C-A, Dantas G
Open Forum Infectious Diseases | 10.1093/ofid/ofx163.245

2015 Autologous Fecal Microbiota Transplantation as a Strategy to Prevent Colonization With Multidrug-Resistant Organisms Following Antimicrobial Therapy
Bulow C, D'Souza A, Hink T, Wallace M, Reske K, Sun X, Kwon JH, Burnham C-A, Dantas G, Dubberke E
Open Forum Infectious Diseases | 10.1093/ofid/ofv133.479

patents

2018 Store-Easy Automotive wheelchair storage device.
Patent US9937088B2
Guertler C, D'Souza AW, Milgrom R, Hong T, Xiao Y, Nadell S

awards

2025-26 Boston Combined Residency Program Fellow Teaching Honor Roll
2025 St. Jude-PIDS Fellowship Award in Basic and Translational Science
2024-25 Boston Combined Residency Program Fellow Teaching Honor Roll
2024-25 Boston Combined Residency Program Equity, Diversity and Inclusion Honor Roll
2022 F. Sessions Cole Award in Pediatrics
2022 Alexander Berg Prize

2022 Washington University Internal Medicine Club Research Award
2021 Gold Humanism Honor Society Inductee
2020 Spencer T. and Ann W. Olin Fellow
2017-18 Burroughs Wellcome Fund, Institutional Program Unifying Population and Laboratory-Based Sciences
2016 BioEnterpreneurship Core Bench to Business Competition
2015 IDEA Labs Demo Day Finalist
2013 Alan S. Tetelman Fellowship for International Research in the Sciences
2013 Paul K. and Evelyn Elizabeth Cook Richter Fellowship
2013 Connecticut Space Grant College Consortium Student Project Grant

2012 CBC Spouses Cheerios Brand Health Initiative Scholarship
2011 CBC Louis Stokes Health Scholarship
2010 CBC Louis Stokes Health Scholarship

posters

2019 Spatiotemporal dynamics of multidrug resistant bacteria on intensive care unit surfaces.
IMM/MICRO Mini WIP

2019 Cotrimoxazole prophylaxis increases resistance gene prevalence and α-diversity but decreases β-diversity in the gut microbiome of HIV-exposed, uninfected infants.
American Society of Microbiology (ASM) Microbe

2019 Longitudinal analysis of ICU surface microbial contamination in the US and Pakistan
NIAID/IDSA Infectious Diseases Research Careers Meeting

2018 Longitudinal Analysis of ICU Surface Microbial Contamination in the US and Pakistan
American Physician Scientist Association (APSA) Midwest meeting

2018 Longitudinal Analysis of ICU Surface Microbial Contamination in the US and Pakistan
Lake Arrowhead Microbial Genomics Conference

2018 Longitudinal Analysis of ICU Surface Microbial Contamination in the US and Pakistan
American Society of Microbiology (ASM) Microbe

2017 Longitudinal Analysis of ICU Surface Microbial Contamination in the US and Pakistan
American Society of Microbiology (ASM) Microbe

2017 Longitudinal Analysis of ICU Surface Microbial Contamination in the US and Pakistan
5th annual Global Health & Infectious Disease Conference

2014 Invasive populations of *Aedes aegypti* in Central California
Yale Ecology and Evolutionary Biology Poster Session

2013 The genetics of foraging behavior in *Aedes aegypti*
Cell Symposia: Genes, Circuits, and Behavior

2012 Mannose receptor signaling linking spleen tyrosine kinase and MAP kinase p38 in human macrophages
The Ohio State University SUCCESS program

2009 Relationship of spinning time and speed to photoresist thickness on semiconductor wafers
Army Research Labs: Science and Engineering Apprentice Program

invited talks

2022 Impacts of swine farm exposure on the human gut microbiota and resistome though genomics and metagenomics
Leman Swine Conference

2021 6 mo M presenting with 5 days of fever, emesis, and irritability
BJC/SLCH Infectious Diseases Grand Rounds

2021 An exploration of Emergency Department turnaround time data
WUSTL Clinical Informatics Seminar

2021 Total thyroidectomy in a 15 yo F with familial adenomatous polyposis
WUSTL GI Seminar

2021 Angiomyolipoma in a 9 yo F with tuberous sclerosis
WUSTL Pathology Seminar

2020 16S rRNA gene sequencing in bacterial endocarditis
WUSTL Lab Medicine Seminar

2019 Effects of antibiotic prophylaxis on the HIV-exposed uninfected infant gut microbiome
WUSTL MSTP Work in Progress Seminar

2018 Bacterial contamination on hospital ICU surfaces in the United States and Pakistan
WUSTL Microbiology Department Seminar

2018 Bacterial contamination on hospital ICU surfaces in the United States and Pakistan
WUSTL Center for Genome Sciences

2018 Bacterial contamination on hospital ICU surfaces in the United States and Pakistan
American Society of Microbiology Micro

2017 Bacterial contamination on hospital ICU surfaces in the United States and Pakistan
CSB-MMMP-PMB Interdepartmental Retreat

2017 Multidrug resistant organisms on hospital surfaces in the United States and Pakistan
Computational and Systems Biology Retreat

2017 Longitudinal analysis of ICU surface microbial contamination in the United States and Pakistan
American Society of Microbiology Micro

2017 Effects of trimethoprim-sulfamethoxazole on the microbiomes of HIV- infants born to HIV+ mothers
WUSTL Center for Genome Sciences

2017 Antibiotic Resistance: a growing public health crisis
WUSTL Institute of Public Health

2013 Drop Team and Rayleigh-Taylor instability
Canyon High School AP biology class

2012 Mannose Receptor Signaling
SUCCESS

teaching

2024-pres. Clinical instruction of residents from the Boston Combined Residency Program
2022-pres. Clinical instruction of medical students from Harvard, Boston University, and Tufts Medical Schools
2017-20 Computational Teaching Assistant | Fundamentals of Biostatistics | Washington University in St. Louis
2017-20 Graduate Mentor | Visiting scholars, rotation students, and undergraduates | Dantas Lab
2017 Teaching Assistant | Microbes and Pathogenesis | Washington University in St. Louis
2017 Graduate Mentor | IPH Summer Program | Washington University in St. Louis
2016 Graduate Mentor | Eureka Program | Washington University in St. Louis
2013-14 Lecturer | Origins of Life | Yale Splash
2011-14 Lecturer | The Science of Ice Cream | Yale Splash

professional memberships

2024-pres. Pediatrics Infectious Diseases Society
2022-pres. American Academy of Pediatrics
2021-pres. Gold Humanism Honor Society
2019-pres. Infectious Diseases Society of America
2016-pres. American Society for Microbiology

activities and leadership

2022-pres. Reviewer — Gut Microbes
Independent peer review of manuscripts on bacterial genomics and metagenomics

2022-pres. Reviewer — Frontiers
Independent peer review of manuscripts on bacterial genomics and metagenomics

2021-pres. Reviewer — Microbiology Spectrum
Independent peer review of manuscripts on bacterial genomics and metagenomics

2020 Volunteer — Maker Task Force, Washington University in St. Louis/Barnes Jewish Hospital
Helped produce face shields for hospital

2019-2020 Volunteer — Saturday Neighborhood Health Clinic
Seeing patients as a medical student volunteer for the free clinic

2014-pres. Yale Admissions Interviewer — Yale Alumni Schools Committee
Interviewer for Yale College admissions of prospective undergraduate students

2015-17 Physicians for Human Rights — Washington University in St. Louis
Co-President of club focusing on human rights and political issues that relate to medicine, including health-care for prisoner, refugees, and underserved communities

2014-17 Store-Easy — Washington University in St. Louis - Sling Health
Company seeking to improve manual wheelchair storage in automobiles

2012-14 NASA Microgravity University (Drop Team) — Yale University
Undergraduate research team selected by NASA to conduct fluid mechanics research in zero-gravity

2012-14 Splash at Yale — Yale University
Undergraduate group teaching short courses to middle and high school students

2011-14 Big Sib — Yale University - Pierson College
Mentor to freshman students at Yale University